INCH POUND

MIL-D-70443 (AR)
AMENDMENT 1
26 August 1992

MILITARY SPECIFICATION

DISPLAY UNIT, SIGHT, HELICOPTER HEAD UP SUBSYSTEM: M76

This amendment forms a part of Military Specification MIL-D-70443 (AR), dated 30 November 1983, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Bottom of page: Add the following distribution statement "Distribution Statement A. Approved for public release; distribution is unlimited."

PAGE 4

3.4.1.1: Delete and substitute

"3.4.1.1 Electrical signal interface. The HUD display unit interface with the signal processor shall be as defined in TABLE I."

PAGE 5

3.4.1.3.2, First sentence: Delete "AN/PVS-5" and substitute "AN/PVS-5 or AN/AVS-6".

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3.4.1.2.1: Delete table portion found on page 5 in its entirety and substitute

Signal Name	Nominal Voltage (±2%)	Maximum Continuous Current	% Regulations 10% to Full Load
Deflection Boost PWR	+40V	0.10A	3%
Deflection Boost PWR	-40V	0.10A	3%
Deflection PWR	+16V	0.8A	3%
Deflection PWR	-16V	0.55A	3%
Z-AMP PWR	+85V	0.04A	3%
Z-AMP PWR	-85V	0.003A	10%
Edge Lit Panel	0 to 28V	0.25A	

PAGE 6

Delete paragraph number 3.4.2.2.2.2 and substitute "3.4.2.2.2".

3.4.2.2.2, line b: Delete "display" and substitute "disparity".

PAGE 7

3.4.2.2.3: Delete in its entirety and substitute

"3.4.2.2.3: Optical Transmissibility. The individual combining glass plates shall have a minimum optical transmissibility of 75 percent required, over the FOV of the HUD.".

PAGE 8

Figure 1, at alternate eye position: Delete "STA 130.00" and substitute "STA 122.00".

Figure 1, at A/C HUD mounting plane: Delete "27.5° REF" and substitute "27.55° REF".

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PAGE 10

Table I, under parameters of deflection X- : Delete "± 10V diff = ± 10.5° deflection (183.259 mr)." and substitute "± 10V diff = ± 10° deflection (174.533 mr).".

Table I, under parameters of deflection Y- : Delete "± 10V diff = ± 10.5° deflection (183.259 mr)." and substitute "± 10V diff = ± 10° deflection (174.533 mr).".

PAGE 11

3.4.2.3.3.3: Delete paragraph in its entirety and substitute:

"3.4.2.3.3.3 Brightness uniformity. For a given brightness setting, the average line brightness of symbols shall not vary more than 40 percent over the FOV of the HUD. Brightness variation is defined as follows:

$$B \text{ variation} = \frac{Bmax - Bmin}{Bmax + Bmin}$$

Where:
Bmax = maximum symbol line brightness
Bmin = minimum symbol line brightness".

PAGE 12

3.4.2.5.5: Delete "AN/PVS-5" and substitute "AN/PVS-5 or AN/AVS-6".

PAGE 13

3.4.2.6.4: Delete "AN/PVS-5" and substitute "AN/PVS-5 or AN/AVS-6".

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PAGE 14

3.4.3.3: Delete in its entirety and substitute

"3.4.3.3 Panel lighting. The instrument shall be provided with integral 28-volt blue-green lighting conforming to MIL-L-85762.

a. Coating. Reflection reducing coating shall be in accordance with MIL-C-14806.

b. Reduced voltage operation. The lighting shall operate satisfactorily at reduced voltage. The relative brightness and color of the display shall be uniform and all markings shall be legible when the lighting is operated at 16.50 ± 0.02 Vdc and viewed from a distance of 28 to 30 inches.".

PAGE 15

3.4.5.1.8: Delete in its entirety.

PAGE 16

3.4.8.2, Electromagnetic interference: Add "CEO3 broadband limits shall be relaxed 23 dB between 1.5 and 5.5 MHz."

PAGE 18

Table II, Line 9: Delete
"9 Rain 3.4.5.1.8 4.6.3.9".

Table III, First line: Delete
"101 Electrical signal interface 3.4.1.1 Appendix A".

PAGE 22

4.6.3.9: Delete in its entirety.

PAGE 23

Following 4.6.3.11: Add new paragraph

"4.6.3.12 Electromagnetic radiation. The subsystem shall be installed in laboratory test fixtures in a manner that will simulate actual service usage and tested to MIL-STD-461 for the requirements of 3.4.8."

PAGE 24

Figure 2, y-axis: Delete "0.36" and substitute "0.036".

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PAGE 26

Appendix A: Delete Appendix A in its entirety.

PAGE 53

Appendix B, under input voltage cycling, step c: Delete "100%" and substitute "110%".

PAGE 55

Appendix B, Figure 6, at top of figure: Delete

"<----28.0V---->|<----30.8V---->|<----25.6V---->"
±.84 ±.92 ±.77

and substitute

"<----28.0V---->|<----30.8V---->|<----25.2V---->"
+1.40 +1.54 +1.26
-0.56 -0.62 -0.50

PAGE 56

Note 3 : Delete in its entirety and substitute

"Note 3 Equipment ON; input voltage (28 VDC) at 100%, 110%, and 90%, respectively, during 1/3 of the total time of heating period plus high temperature period; momentarily off between input voltage changes.".

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